

AYE
FIT HUB

• AYE FIT HUB • DAY 16 OF 30

LOWER BODY FLEXIBILITY

Hips · Hamstrings · IT Band · Adductors · Piriformis

Day 16 of 30 | CRF Flexibility | AYE Fit Hub

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- Anatomy of tightness — why hips, hamstrings & IT band become restricted
- Hip flexor release — 6 stretches with holds, sets and execution cues
- Hamstring protocol — beginner to advanced PNF with 7 progressions
- IT Band — what actually works vs what does not (evidence table)
- Adductor & piriformis stretches — inner thigh and deep glute release
- Complete 30-minute lower body flexibility routine with full timing table
- Stretch science — muscle spindle reflex, GTO, PNF autogenic inhibition
- 6-week progression plan + flexibility benchmarks to track your progress

Why Lower Body Flexibility Is Your Foundation

Most athletes focus entirely on strength and cardio — and completely ignore flexibility. That is a costly mistake. **Tight hip flexors cause lower back pain.** Tight hamstrings increase injury risk during sprinting, jumping, and deadlifts. A restricted IT band produces the burning lateral knee pain that ends running careers prematurely. These are not minor inconveniences — they are structural problems that compound over months and years of neglect.

Every hour of sitting shortens your hip flexors, compresses your lumbar discs, and deactivates your glutes. The average office worker sits 10–14 hours per day. Physical therapists call the result **Lower Crossed Syndrome**: tight hip flexors and hamstrings pull the pelvis forward into an exaggerated anterior tilt, causing chronic low back pain, poor posture, and reduced athletic performance. Day 16 is your direct, systematic counter-attack against all of it.

Today we cover all five major lower body flexibility zones — **hip flexors, hamstrings, IT band, adductors, and piriformis** — with specific protocols, hold times, progressions, and the science behind why each approach works. By the end you will have a complete 30-minute routine you can run any day of the week.

The 5 Lower Body Muscle Groups Targeted Today

Lower Body Flexibility Targets — 5 Key Muscle Groups

HIP FLEXORS	HAMSTRINGS	IT BAND	ADDUCTORS	PIRIFORMIS
<i>Front of hip</i>	<i>Back of thigh</i>	<i>Outer thigh</i>	<i>Inner thigh</i>	<i>Deep glute</i>
Iliopsoas Rectus Femoris	Biceps Femoris Semitendin.	Iliotibial Band (fascia)	Add. Magnus Gracilis	Piriformis (sciatic nerve)
<i>Critical for desk workers</i>	<i>Most commonly tight muscle</i>	<i>Runners & cyclists</i>	<i>Limits squat depth if tight</i>	<i>Sciatic pain from sitting</i>

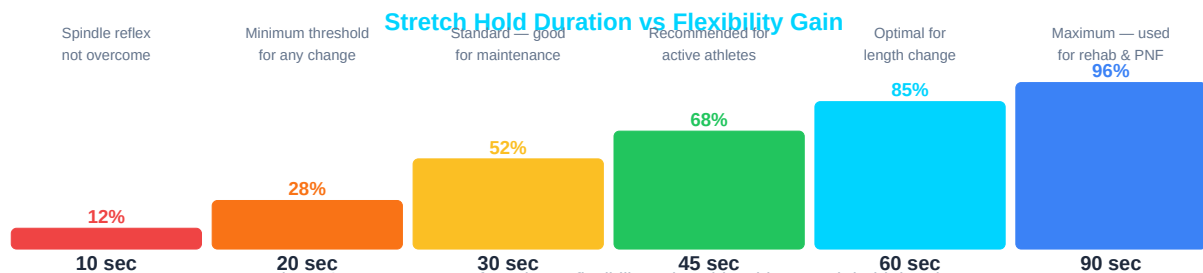
Figure 1 — Each group requires a different stretch technique and hold protocol

Muscle Group	Primary Action	Tightness Symptom	Who Is Most Affected
Hip Flexors	Flex hip, lift knee	Low back arch, anterior pelvic tilt, hip pain	Office workers, cyclists, drivers
Hamstrings	Extend hip, flex knee	Posterior pelvic tilt, pulled hamstring, knee pain	All athletes, runners especially
IT Band (fascia)	Stabilise lateral knee	Burning outer knee pain at 30–40 min of running	Runners, cyclists, hikers
Adductors	Adduct hip (close legs)	Groin tightness, limited squat depth, knee valgus	Footballers, team sport athletes
Piriformis	Externally rotate hip	Deep glute ache, sciatic nerve referral to leg	People sitting 6+ hours per day
Hip Rotators	Internal/external rotation	Limited squat mobility, knee cave on landing	All strength and power athletes

How Long Should You Hold a Stretch?

Hold duration is the single most misunderstood variable in flexibility training. The research is clear: **10-second holds produce almost no lasting change.** The muscle spindle reflex — a protective neurological mechanism

— keeps the muscle contracted for the first 20 seconds of any stretch. Only after that does the Golgi tendon organ (GTO) begin to override the reflex and allow the muscle to genuinely relax and elongate.



BREATHING TECHNIQUE — On every exhale, consciously relax and deepen into the stretch by 2–5%. Inhale: hold position. Exhale: release and sink further. This alone can double your flexibility gains compared to stretching while holding your breath, because slow exhalation activates the parasympathetic nervous system — reducing protective muscle tone throughout your body.

Hip Flexor Release — The Most Critical Zone

The hip flexors are the most chronically shortened muscle group in modern adults. The **iliopsoas** (iliacus + psoas major) runs from your lumbar vertebrae L1–L5 through your pelvis and attaches to the lesser trochanter of the femur. When shortened, it pulls your lumbar spine forward and compresses your lower discs. The **rectus femoris** crosses both the hip and the knee — meaning it becomes doubly restricted by both sitting and leg extension positions. Both must be targeted separately for complete release.

Golden Rule: Hold each stretch for a minimum of 45 seconds to begin neurological relaxation. For meaningful length change you need 60–90 seconds per side, per stretch, at least 5 days per week for 4–6 consecutive weeks. Short holds are better than nothing but will not produce lasting structural change.

Stretch Name	Starting Position	Key Execution Cue	Hold / Sets	Muscle Target
Kneeling Hip Flexor Lunge	Half-kneeling, back knee down	Tuck pelvis under (posterior tilt), lean forward	45–60s / 2/side	Iliopsoas (deep)
Standing Quad Stretch	Standing, hold back foot	Drive hip forward — do not arch lumbar	30–45s / 2/side	Rectus Femoris
Couch Stretch	Back foot elevated on bench/wall	Knee close to wall, torso fully upright	60–90s / 2/side	Rectus Femoris (intense)
Low Lunge with Rotation	Deep lunge, back knee down	Rotate torso toward front leg, reach arm up	30–45s / 2/side	Hip flexor + thoracic
90/90 Hip Flexor	Seated on floor, both knees 90°	Sit tall, lean forward from hip over front leg	45–60s / 2/side	Iliopsoas + hip capsule
Psoas Release (Supine)	Lie flat, pull one knee to chest	Keep opposite leg completely flat on floor	45–60s / 2/side	Psoas major (passive)

Self-Assessment: Thomas Test for Hip Flexor Tightness

The Thomas Test is the gold-standard physiotherapy assessment for hip flexor restriction. You can perform a modified version at home to gauge your starting point and track progress over your 6-week flexibility programme:

Assessment Test	How to Perform	Result — What It Means
Modified Thomas Test	Lie at edge of bed, pull one knee firmly to chest — observe opposite leg	Leg stays up = tight iliopsoas; Knee bends >90° = tight rectus femoris
Prone Hip Extension	Lie face down, lift one straight leg off floor	Leg lifts <15° without arching back = very restricted psoas major
Standing Anterior Tilt	Stand sideways in mirror — is your pelvis tilted forward?	Visible anterior tilt = shortened hip flexors pulling pelvis forward
Kneeling Lunge Comfort	Hold kneeling lunge 60s — any lower back sensation?	Back arching or pain = hip flexors pulling spine into extension

PROGRESSION RULE — Start with the kneeling lunge for 2 weeks. Once you hold 60 seconds with zero lumbar discomfort and a fully tucked pelvis, progress to the couch stretch. Once the couch stretch is comfortable at 90 seconds, your hip flexor mobility is excellent.

Hamstring Flexibility — The Most Misunderstood Stretch

The hamstrings are three muscles running down the back of your thigh: **biceps femoris** (outer head), **semitendinosus**, and **semimembranosus** (inner). Together they flex the knee and extend the hip. Tight hamstrings tilt the pelvis posteriorly — the opposite problem to tight hip flexors — flattening the lumbar spine and increasing disc shear stress.

The most common mistake: people round their lower back during a forward fold and believe they are stretching their hamstrings. They are not. A rounded lumbar spine means you are stretching your lumbar fascia and spine — not your hamstrings. The key is to maintain a neutral pelvis with a flat back ("proud chest") before folding forward. If you cannot fold forward with a flat back, you are not yet ready for standing forward folds — begin with supine work.

Level	Stretch Name	How to Perform	Hold / Sets	Progress Marker
BEGINNER	Supine Strap Stretch	Lie flat, loop towel/band around foot, lift straight leg to gentle tension — back flat	45s × 2/side	Reach 70° hip flexion, back flat
BEGINNER	Seated Chair Stretch	Sit at chair edge, straighten one leg, hinge forward from hip with flat back	30–45s × 2/side	Hands reach shin, no lumbar rounding
INTERMEDIATE	Standing Forward Fold	Hinge at hips with slight knee bend, flat back — let hands hang, DO NOT round	45s × 3	Hands below knee with flat back
INTERMEDIATE	Downward Dog Hold	Press palms and heels down, straighten knees gradually, one heel at a time	30s × 4	Both heels comfortably on floor

INTERMEDIATE	Single-Leg RDL Stretch	One leg behind on bench, hinge over standing leg, pelvis square	45s × 2/side	No back sensation — only hamstring pull
ADVANCED	PNF Contract-Relax	Lift leg to end range, contract hamstring 6s (push against strap), relax and deepen	6s + 30s × 3/side	2–5° ROM gain per session
ADVANCED	Active Straight Leg Raise	Lie flat, actively kick straight leg toward ceiling using hip flexors	10 reps × 3/side	Leg reaches 80°+ without pelvis lifting

PNF Stretching — The Most Effective Method

PNF (Proprioceptive Neuromuscular Facilitation) consistently outperforms static stretching in research, achieving 2–4 times more range of motion gain per session. It works by exploiting two neurological mechanisms simultaneously: **autogenic inhibition** (GTO relaxation after contraction) and **reciprocal inhibition** (contracting the agonist causes the antagonist to relax). The cycle below is the standard clinical protocol used by physiotherapists:

PNF Stretching Cycle — Contract-Relax Method

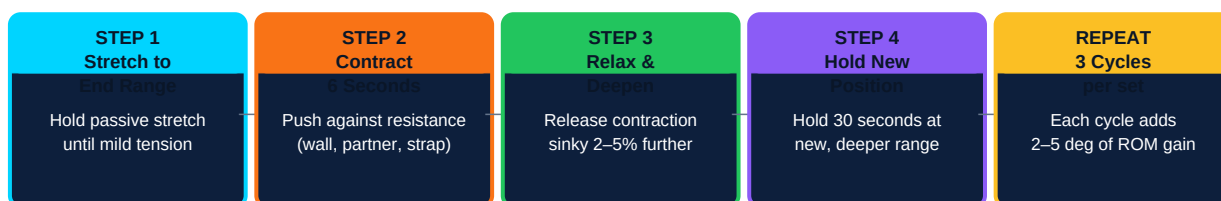


Figure 3 — PNF achieves 2–4× more ROM gain per session than static stretching alone

IT Band — What It Is and What Actually Works

The **iliotibial band (IT band)** is not a muscle — it is a thick band of connective tissue (fascia) running from the hip (tensor fasciae latae and gluteus maximus) along the outer thigh to the lateral tibial condyle just below the knee. Because it is dense fascia and not contractile tissue, it cannot be "stretched" the same way a muscle can. Traditional IT band stretches (crossing one leg behind in standing) have very limited clinical evidence. What actually produces results is a combination of **foam rolling the surrounding musculature, releasing the TFL and glute max**, and critically — **strengthening hip abductors** to reduce the lateral loading that caused the problem in the first place.

IT Band Approach	Duration / Sets	Evidence	What It Does
Foam roll TFL (upper outer hip)	60–90s × 2 sides	HIGH	Releases the TFL muscle that pulls the IT band taut — most effective IT band treatment
Foam roll outer quad + glute max	60s each × 2 sides	HIGH	Reduces fascial tension in all tissue feeding into the IT band proximally and distally
Hip crossover stretch (standing)	30–45s × 2 sides	MODERATE	Mild IT band tractioning — useful but limited standalone effect

Side-lying IT band stretch	30–45s × 2 sides	LOW–MODERATE	Limited effect — fascia does not elongate meaningfully like muscle belly tissue
Clamshell hip abduction	3 × 15 reps	HIGH	Strengthens glute medius — removes the root biomechanical cause of ITBS in runners
Single-leg glute bridge	3 × 12 reps	HIGH	Activates glute max — reduces TFL compensation and lateral hip overload during gait

Adductor Flexibility — The Forgotten Inner Thigh

The adductors (adductor magnus, longus, brevis, gracilis, and pectineus) run from the inner pelvis and pubic bone to the inner femur and tibia. Tight adductors limit squat depth, reduce hip abduction range, and contribute to **knee valgus** — knees caving inward during squats, jumps and landings. This is a significant **ACL injury risk factor**, particularly in female athletes. Despite this, adductor flexibility is the most consistently neglected flexibility zone across recreational and competitive training programmes.

Stretch Name	Starting Position	Key Cue	Hold / Sets
Sumo Squat Hold	Wide stance, toes 45° out, sink deep	Elbows push knees wide, chest tall, heels flat	45–60s / 3 sets
Side Lunge Adductor	Wide stance, shift weight to one side	Hinge toward bent knee from hip, keep opposite leg straight	30–45s / 2/side
Butterfly (Cobbler Pose)	Seated, soles together, knees out	Gently press knees toward floor — never bounce	45–60s / 2 sets
Frog Stretch	Hands and knees, widen knees progressively	Sink hips slowly toward floor, toes pointing outward	60s / 2 sets
Standing Wall Adductor	Stand beside wall, slide foot up wall laterally	Keep hips square, lean slightly away from wall	30s / 2/side

Piriformis Release — Deep Glute and Sciatic Relief

The **piriformis** is a small, deep external rotator of the hip located directly beneath the gluteus maximus, running from the sacrum to the greater trochanter of the femur. In approximately 17% of the population, the sciatic nerve passes directly through the piriformis rather than beneath it. When the piriformis tightens — from prolonged sitting, hip weakness, or compensatory overuse — it can compress the sciatic nerve, producing burning, shooting, or tingling pain running from the deep glute down the leg. This is called **piriformis syndrome** and is frequently misdiagnosed as lumbar disc sciatica.

NERVE CAUTION — If stretching the piriformis produces sharp shooting pain, tingling, or numbness below the knee, stop immediately. Aggressive stretching can worsen true nerve irritation. Consult a physiotherapist to differentiate piriformis syndrome from lumbar disc pathology before continuing.

Stretch Name	How to Perform	Hold	Intensity
Figure-4 Supine	Lie on back, cross ankle over opposite knee, pull that thigh to chest with both hands	45–60s / 2/side	Low — best starting point

Seated Figure-4	Sit in chair, cross ankle over knee, sit tall and lean forward from hips	45–60s / 2/side	Low — do at desk, every hour
Pigeon Pose	From plank, bring shin forward parallel to front of mat, fold torso over it	60–90s / 2/side	High — deep comprehensive stretch
90/90 Hip Stretch	Both knees at 90° on floor (front and back), sit upright or lean forward	60s / 2/side	High — best overall hip rotator
Supine Cross-Body Pull	Lie flat, pull bent knee across body toward opposite shoulder	30–45s / 2/side	Moderate — targets piriformis + TFL
Standing Figure-4 Hold	Stand on one leg, cross other ankle over thigh, sit back into a one-leg squat	30–45s / 2/side	Moderate + balance challenge

Your Complete 30-Minute Routine

Complete 30-Minute Lower Body Flexibility Routine — Block Timeline



Order	Exercise / Stretch	Time	Muscle Targeted	Key Cue
1	Light march or leg swings	5 min	Full lower body warm-up	Never stretch cold — warm tissue is 37% more extensible
2	Kneeling hip flexor lunge	45s×2/side	Iliopsoas	Tuck pelvis, no lumbar arch, feel stretch front of hip
3	Couch stretch (or standing quad)	60s×2/side	Rectus femoris	Knee near wall, torso fully upright
4	Supine hamstring with strap	45s×2/side	Hamstrings	Flat back, opposite leg fully extended on floor
5	Standing forward fold	45s×2	Hamstrings	Hinge from hips, proud chest, slight knee bend
6	Foam roll TFL (outer hip area)	60s/side	TFL — IT band relief	Slow 1 inch/second — pause on tight spots
7	Hip crossover IT band stretch	30s×2/side	IT band + TFL	Cross leg behind, mild lateral tractioning only
8	Butterfly pose	60s×2	Adductors (inner thigh)	Gently press knees toward floor, never bounce

9	Frog stretch	60s×2	Adductors (deep)	Slowly widen knees, relax hips toward floor
10	Figure-4 piriformis (supine)	60s×2/side	Piriformis	Pull thigh gently, keep lower back flat on floor

The Neuroscience of Flexibility

Flexibility is primarily a **neurological phenomenon**, not a structural one. Your muscles are not physically short — your nervous system is being overprotective. When a muscle is stretched rapidly, **muscle spindles** (intrafusal muscle fibres) fire a reflex contraction to protect the joint — the myotatic reflex. Holding a stretch slowly and progressively allows the **Golgi tendon organ (GTO)** — located at the muscle-tendon junction — to override this reflex via **autogenic inhibition** after approximately 20–30 seconds. The muscle genuinely relaxes, and you can deepen the stretch. This is why hold duration matters so much.

Mechanism	Trigger	Physiological Result	Practical Application
Muscle Spindle Reflex	Rapid or forceful stretch	Protective contraction (resists stretch)	Hold slowly — never bounce, never force
GTO Autogenic Inhibition	Sustained tension >20s	Spindle overridden — muscle relaxes	Hold 30–90s to access genuine ROM
Reciprocal Inhibition	Agonist contraction	Antagonist automatically relaxes	PNF — contract hamstring to relax it
Fascial Creep	Low-load sustained stretch	Collagen fibres gradually elongate	Daily stretching — changes take 4–8 weeks
Neural Drive Reduction	Repeated daily stretching	Nervous system reduces protective tone	Frequency matters — 5–7×/week minimum
Thermal Effect	Elevated tissue temperature	Collagen extensibility increases 37%	Always stretch post-workout when warm

Common Mistakes That Block Flexibility Progress

Mistake	Root Cause	Correct Approach
Bouncing during stretches	Using momentum, rushing the session	NEVER bounce — activates the spindle reflex, causes micro-tears in the musculotendinous junction
Stretching cold muscles	Skipping warm-up, going straight in	Always do 5 min of light movement first — cold collagen is stiff and injury-prone
Holding under 20 seconds	Believing short holds "count"	Minimum 30s for maintenance; 60s for change; 90s for advanced or rehab work
Rounding back on hamstring stretch	Confusing lumbar flex with hamstring	Hinge from hips with a flat, proud back — feel the stretch in the thigh, not the lower back
Only stretching the IT band directly	Treating fascia like muscle	Foam roll TFL + strengthen glute med — the IT band cannot elongate like muscle tissue

Irregular, infrequent stretching	No structured routine	Minimum 5×/week; daily is ideal — collagen remodelling needs repeated consistent stimulus
Expecting fast results	Unrealistic timeline	Meaningful flexibility change takes 4–8 weeks; measure with benchmarks, not feelings
Stretching through sharp pain	Misapplying "no pain no gain"	Stretching should feel like mild-to-moderate tension only — sharp pain is a stop signal

6-Week Lower Body Flexibility Progression

Week	Daily Focus	Session Length	Priority Stretches	Goal to Achieve
1–2	Hip flexors + basic hamstrings	15–20 min	Kneeling lunge, supine strap hamstring	Build habit; reduce anterior pelvic tilt; hold lunge 45s
3–4	Full lower body routine	20–25 min	Full hip/hamstring + butterfly + IT foam roll	All 5 zones covered daily; hold each 45–60s comfortably
5	Add piriformis + PNF hamstring	25–30 min	Figure-4 pigeon, PNF contract-relax method	Begin PNF — aim for 2–5° ROM gain per session
6	Full 30-min routine + retest	30 min	Complete 10-step programme	Re-test Thomas Test + forward fold — compare to Week 1

Flexibility Benchmarks — Know Your Level

Assessment Test	Beginner	Intermediate	Advanced Goal
Seated forward fold	Hands reach knees only	Hands reach ankles	Flat palms on floor, knees straight
Standing forward fold (flat back)	Hands mid-shin	Hands at ankle level	Palms flat on floor, back neutral
Hip flexor lunge (60s test)	Cannot hold 30s upright	Holds 45s with slight discomfort	60s each side, zero back sensation
Butterfly (knee-to-floor distance)	Knees 45°+ from floor	Knees 20° from floor	Both knees touching floor
Thomas Test	Leg stays well above table	Leg level with table	Leg drops clearly below table level
Piriformis (figure-4 hip angle)	Cannot cross ankle over knee	Can cross, minimal hip opening	Full figure-4, hips square, lean forward easily

Day 17 Preview: Protein and Muscle — How Much, When, and Why. Tomorrow we move from flexibility to nutrition and answer the most hotly debated question in fitness science: exactly how much protein do

you need to build and maintain muscle, what timing matters, and which sources deliver the best results for your training goals.